ANALYSTLAB AFRICA

Data Science Internship Programme

Data Preprocessing Report

Week 2: Feature Engineering and Data Preprocessing for Machine Learning

Document control

Programme	Data Science Internship
Week	Week 2
Project	Data preprocessing for Machine Learning: prediction of heart disease

1. Context and overview of preprocessing

This report documents the preprocessing decisions applied to the **Heart Failure Prediction** dataset (Kaggle, fedesoriano), selected from the two options of the assignment (Option 1). The objective is to transform raw medical data into a dataset ready for Machine Learning, in order to predict the **presence (1) or absence (0) of heart disease** in a patient. The target variable is **HeartDisease**.

The raw dataset contains **918 observations** and **12 columns**: 11 explanatory variables and the target variable.

Type	Variables
Continuous numerical	Age, RestingBP, Cholesterol, MaxHR, Oldpeak
Binary coded 0/1	FastingBS, HeartDisease (target)
Categorical (text)	Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope

The following sections follow the structure required by the assignment. The chronological order of execution in the notebook is summarised below.

No.	Step	Technique	Result
1	Inspection	isnull(), duplicated(), info()	0 missing values, 0 duplicates
2	Data type correction	FastingBS: 0/1 → No/Yes; HeartDisease cast to integer	Consistent types
3	Outliers	IQR method (1.5 × IQR), removal	918 → 702 rows
4	Cleaned export	to_csv()	HeartDisease_clean.csv (702 × 12)
5	Encoding	One-Hot Encoding (get_dummies, drop_first)	15 explanatory variables
6	Variable selection	Correlations, heatmap, χ^2, Tschuprow's T	Variables ranked by relevance
7	Train/test split	Stratified train_test_split 80/20, random_state = 42	561 train / 141 test
8	Feature scaling	StandardScaler on 5 numerical variables	Mean 0, standard deviation 1 (train)
9	Final export	to_csv()	HeartDisease_ML.csv (702 × 17)

2. Data cleaning

2.1 Handling missing values

The instruction df.isnull().sum() returns **0 missing values for all 12 columns**. No imputation was therefore necessary at this stage.

However, the absence of null values does not guarantee the absence of invalid values. The outlier detection (section 5) made it possible to discard physiologically implausible measurements, in particular cholesterol levels equal to 0 (the lower IQR threshold is 32.6 mg/dl). These values most likely correspond to missing data coded as 0 in the source.

2.2 Removing duplicates

df.duplicated().sum() returns **0**: no duplicate record was detected, so no row was removed at this step.

2.3 Correcting data types

- **Numerical variables** (Age, RestingBP, Cholesterol, MaxHR: integers; Oldpeak: decimal): types are correct, no conversion required.
- **FastingBS** (fasting blood sugar > 120 mg/dl): converted from 0/1 to **No/Yes** to make the charts and their interpretation easier to read. The variable is then re-encoded as an indicator variable (section 3).
- **HeartDisease**: explicit conversion to integer. The target variable is deliberately kept in numerical 0/1 form, the format expected by classification algorithms.
- **Text variables** (Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope): treated as nominal categorical variables and encoded in section 3.

The original column names (English, explicit) were kept in order to preserve the correspondence with the source's data dictionary.

3. Variable encoding

3.1 Technique applied: One-Hot Encoding

The six categorical variables were transformed into indicator variables (0/1) using pd.get_dummies(..., drop_first=True): Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope and FastingBS.

Original variable	Categories	Columns created	Dropped reference
Sex	M, F	Sex_M	F
ChestPainType	ASY, ATA, NAP, TA	ChestPainType_ATA, _NAP, _TA	ASY
RestingECG	Normal, ST, LVH	RestingECG_Normal, _ST	LVH
ExerciseAngina	N, Y	ExerciseAngina_Y	N
ST_Slope	Up, Flat, Down	ST_Slope_Flat, _Up	Down
FastingBS	No, Yes	FastingBS_Yes	No

In total, the encoding produces **10 indicator columns** which are added to the 5 numerical variables, giving **15 explanatory variables**.

3.2 Justification of the choice

- **Nominal variables without a natural order.** Label Encoding would assign arbitrary integers (for example ASY = 0 < ATA = 1 < NAP = 2 < TA = 3). Linear or distance-based models would wrongly interpret this order as a quantity. One-Hot Encoding avoids this bias.
- **Low cardinality** (2 to 4 categories per variable): the increase in the number of columns remains limited (+10).
- **drop_first=True** removes one reference category per variable. This avoids the dummy variable trap (perfect collinearity between the categories of the same variable).
- **Label Encoding not used.** The target HeartDisease is already coded 0/1, and the binary variables are covered by One-Hot encoding with drop_first, which is equivalent to 0/1 coding.

4. Feature scaling

4.1 Variables concerned

The five continuous numerical variables have very different orders of magnitude (cholesterol is around 240, whereas Oldpeak is below 4). All of them were standardised. The indicator variables (0/1) were not scaled.

Variable	Mean	Standard deviation	Minimum	Maximum
Age	52.72	9.54	28	77
RestingBP	131.56	15.42	92	170
Cholesterol	239.71	50.68	85	404
MaxHR	140.55	24.36	71	202
Oldpeak	0.83	0.96	-0.1	3.6

Descriptive statistics of the numerical variables after outlier treatment (702 observations).

4.2 Chosen method: StandardScaler

The transformation applied is z = (x − mean) / standard deviation. Each variable is thus centred (mean 0) and scaled (standard deviation 1).

- **Relevance for the targeted models.** Logistic regression, SVMs, k-nearest neighbours and neural networks are sensitive to the scale of variables. Without standardisation, a variable such as Cholesterol would dominate Oldpeak simply because of its unit.
- **Suitable distributions.** After outlier treatment, the variables no longer contain extreme values that would distort the mean and standard deviation. A RobustScaler is therefore less necessary, and a MinMaxScaler would be more sensitive to boundary values.
- **No data leakage.** The dataset was first split into train (80%, i.e. 561 rows) and test (20%, i.e. 141 rows), stratified on HeartDisease (random_state = 42). The scaler was fitted **only on the training set** (fit_transform), then applied to the test set (transform) with the same parameters.

5. Outlier detection and treatment

5.1 Approach: IQR method

For each numerical variable, the bounds were computed on the original dataset: lower bound = Q1 − 1.5 × IQR and upper bound = Q3 + 1.5 × IQR. Any value outside these bounds is considered an outlier.

Variable	Lower threshold	Upper threshold	Outliers
Age	27.5	79.5	0
RestingBP	90.0	170.0	28
Cholesterol	32.6	407.6	183
MaxHR	66.0	210.0	2
Oldpeak	-2.25	3.75	16

The binary target variable was included in the detection loop with no effect (0 outliers).

5.2 Treatment

Rows containing at least one outlier were **removed**: **216 rows out of 918** (23.5%), leaving **702 observations**. The total per variable (229) exceeds 216 because some rows are outliers on several variables. A check with the original thresholds confirms **0 remaining outliers** for all variables.

Box plots were drawn by HeartDisease class after treatment (extreme values hidden in the display).

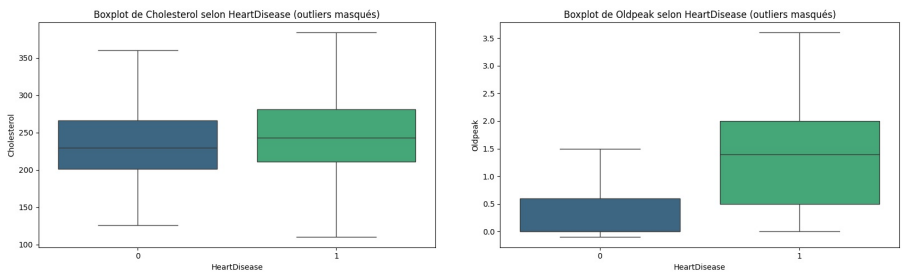


Figure 1 – Box plots of Cholesterol and Oldpeak by HeartDisease (after treatment)

5.3 Justification and limitations

- **IQR method chosen:** it does not assume any normal distribution and remains robust, because quartiles are little

affected by extreme values. The Z-score was not used.

- **Removal rather than correction:** a simple approach that guarantees a dataset without extreme values before scaling.
- **Limitation:** almost a quarter of the patients are excluded, mainly because of Cholesterol (183 values). An alternative would be to replace cholesterol values equal to 0 with the median and to remove only the remaining outliers, in order to retain more data.

6. Feature selection

6.1 Correlation heatmap

The correlation matrix was computed on the numerical variables and the indicator variables, target included.

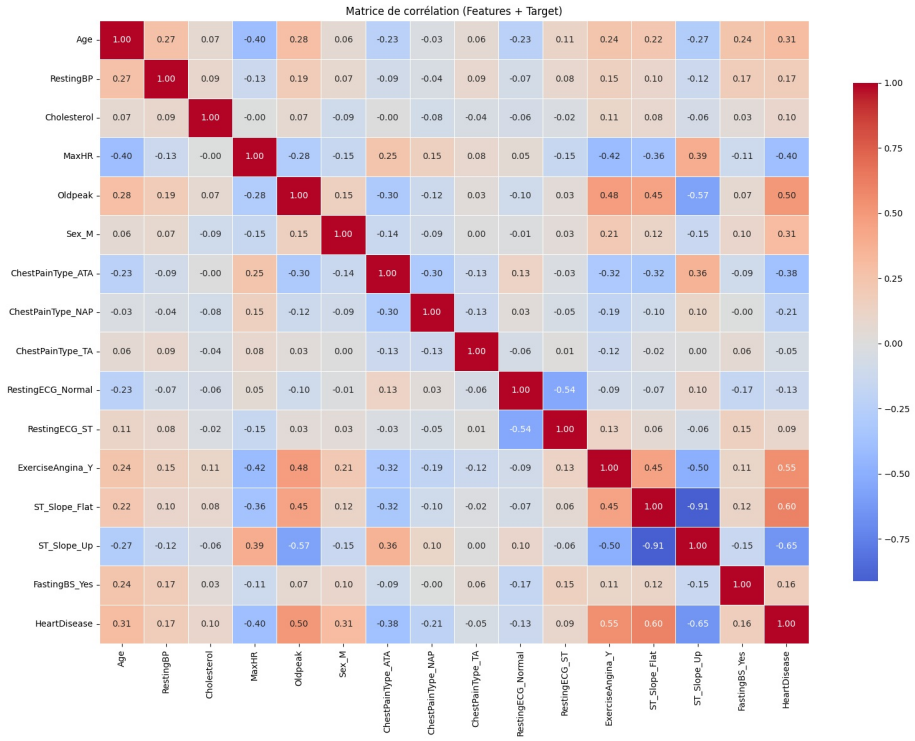


Figure 2 – Correlation matrix (explanatory variables and HeartDisease)

6.2 Most important variables

Variables were ranked by their absolute correlation (maximum for encoded variables) with HeartDisease.

Rank	Variable	Absolute correlation	Rank	Variable	Absolute correlation
1	ST_Slope_Up	0.65	9	ChestPainType_NAP	0.21
2	ST_Slope_Flat	0.60	10	RestingBP	0.17
3	ExerciseAngina_Y	0.55	11	FastingBS_Yes	0.16
4	Oldpeak	0.50	12	RestingECG_Normal	0.13
5	MaxHR	0.40	13	Cholesterol	0.10
6	ChestPainType_ATA	0.38	14	RestingECG_ST	0.09
7	Sex_M	0.31	15	ChestPainType_TA	0.05
8	Age	0.31			

Four columns exceed the threshold of 0.5 (ST_Slope_Up, ST_Slope_Flat, ExerciseAngina_Y and Oldpeak), which correspond to three original variables: **ST_Slope** (slope of the ST segment), **ExerciseAngina** (exercise-induced angina) and **Oldpeak** (ST depression). They are followed by MaxHR, ChestPainType, Sex and Age.

6.3 Statistical validation tests

Two association tests with the target complemented the analysis (cleaned dataset of 702 rows).

Variable	Test	Statistic	p-value
ExerciseAngina	χ^2 (2 categories)	210.33	1.2×10^{-47}
Sex	χ^2 (2 categories)	64.52	9.5×10^{-16}
FastingBS	χ^2 (2 categories)	18.09	2.1×10^{-5}
ST_Slope	Tschuprow's T (3 categories)	0.546	5.3×10^{-65}
ChestPainType	Tschuprow's T (4 categories)	0.410	4.7×10^{-44}
RestingECG	Tschuprow's T (3 categories)	0.112	2.1×10^{-3}

All associations are statistically significant ($p < 0.01$). Their strength, however, is very uneven: **ST_Slope** (T = 0.55) and **ChestPainType** (T = 0.41) are strongly linked to the target, whereas **RestingECG** (T = 0.11) is weakly linked.

6.4 Highly correlated variables and redundancy

- **ST_Slope_Flat / ST_Slope_Up: −0.91.** This correlation is expected, because both columns come from the same variable (the third category, Down, is the reference). It does not constitute redundancy between distinct variables.
- **RestingECG_Normal / RestingECG_ST: −0.54**, for the same reason.
- Between distinct variables, correlations remain moderate: Oldpeak / ST_Slope_Up (−0.57), ExerciseAngina_Y / ST_Slope_Up (−0.50), Oldpeak / ExerciseAngina_Y (0.48), ExerciseAngina_Y / MaxHR (−0.42) and Age / MaxHR (−0.40).

No pair of distinct variables exceeds 0.6 in absolute value: there is **no worrying multicollinearity** justifying the removal of a variable.

6.5 Variables to remove

In the current version, **all 15 explanatory variables were kept** in the final dataset. A weak linear correlation does not imply that a variable is useless, since it may contribute through interactions with other variables. The variables least related to the target are **candidates for removal** to be tested during modelling: Cholesterol (0.10), RestingECG (T = 0.11) and ChestPainType_TA (0.05).

7. Final dataset for Machine Learning

The file **HeartDisease_ML.csv** brings together the standardised train and test sets in their original order.

Characteristic	Value
Dimensions	702 rows × 17 columns
Standardised numerical variables (5)	Age, RestingBP, Cholesterol, MaxHR, Oldpeak
Indicator variables (10)	Sex_M, ChestPainType_ATA / _NAP / _TA, RestingECG_Normal / _ST, ExerciseAngina_Y, ST_Slope_Flat / _Up, FastingBS_Yes
Target	HeartDisease (0 = absence, 1 = presence)
Identification column	Set (train / test): to be excluded from the explanatory variables during modelling
Other file produced	HeartDisease_clean.csv: 702 × 12, readable variables, not standardised

> **Is the dataset ready for Machine Learning? Yes.** It contains no missing values, no duplicates and no outliers according to the IQR criterion; all variables are numerical; the continuous variables are standardised without data leakage; and the train/test split is stratified.